

To do:

- Revise result to have smooth story about temporal and spatial patterns

Title: Spatial and Temporal Changes in Fish Assemblages of the Upper Mississippi River

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Journal Target:

Environmental Biology of Fishes

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ACKNOWLEDGMENTS

We thank the Upper Mississippi River Restoration Program's Long Term Resource Monitoring affiliated scientists that have developed and maintained a high-quality monitoring program, and the individual biologists who have collected the fisheries data that this work relied upon. We appreciate the internal reviewers at the U.S. Geological Survey for improving the manuscript. Any use of trade, firm, or product names is for descriptive purposes only and does not imply endorsement by the U.S. Government.

STATEMENTS AND DECLARATIONS

Funding

The data collection and authors DL and KB were funded by the Upper Mississippi River Restoration Program. RAE and DG-R received funding from the U.S. Geological Survey Biological Threats and Invasive Species Research Program.

Competing interests

The authors have no competing interests to declare.

Author contributions

This paper grew out of the Mathematical Biology Working group at the University of Wisconsin—La Crosse that also includes U.S. Geological Survey scientists. Specific author contributions include:

28 JP contributed to Conceptualization, Software, Data Analysis, Methodology, and Visualization
29 KB contributed to Conceptualization, Interpretation, Writing, and Editing.
30 DE contributed to Conceptualization, Software, Data Analysis, Methodology, and Visualization.
31 RAE contributed to Conceptualization, Methodology, and Editing.
32 DG-R
33 DML contributed to Conceptualization, Data analysis, Interpretation, Writing, and Editing.
34 MM
35 GS
36 DS contributed to Conceptualization, Interpretation, and Editing

37

38 **Data availability**

39 All data were publicly accessed on 1 November 2025 through the Upper Mississippi River
40 Restoration Program's Long-term Resource Monitoring data hub ([https://umesc.usgs.gov/ltrm-](https://umesc.usgs.gov/ltrm-home.html)
41 [home.html](https://umesc.usgs.gov/ltrm-home.html)). Data analysis scripts and our secondary datasets are publicly shared on GitHub
42 (https://github.com/jppeirce/ltrm_fish).

43 **Ethics Statement:**

44 All fish were collected under respective state permits. Fish were handled in accordance with
45 American Fisheries Society guidelines for the use of fishes in research:
46 <https://fisheries.org/policy-media/science-guidelines/guidelines-for-the-use-of-fishes-in-research/>

47 **ABSTRACT (250 words)**

48 Freshwater fish assemblages reflect the interplay of environmental conditions, habitat suitability,
49 and species interactions. In large river ecosystems, long-term changes in flow, water quality, and
50 habitat can alter which species co-occur and how assemblages shift over time. The Upper

Mississippi River (UMR) in the United States of America has experienced long-term ecological changes, including habitat modifications, improvements in water quality, and reductions in certain invasive species. These riverine changes have likely influenced the composition and distribution of native fish communities. We used more than three decades of standardized monitoring data collected across four study reaches within a 400-kilometer section of the UMR. This long-term data allowed us to quantify fish assemblages during autumn, a season that reflects conditions prior to overwintering. We applied hierarchical clustering to long-term abundance data to identify meaningful patterns in fish species co-occurrence. Clustering revealed eight ecologically meaningful fish assemblages that exhibited distinct spatial distributions and temporal trends over the 30-year record, indicating pronounced, system-wide shifts in community structure. This approach offers a practical framework for linking assemblage dynamics to habitat change or management actions and supports targeted restoration and conservation strategies aimed at sustaining biodiversity and recreational fisheries in one of North America's largest river systems. Integrating long-term datasets with modern analytical tools advances our understanding of how freshwater fish communities assemble and respond to environmental change.

KEYWORDS: fish assemblages, cluster analysis, temporal patterns, Upper Mississippi river

INTRODUCTION

Local fish assemblages are organized in nonrandom spatial and temporal patterns shaped by environmental filters and biotic interactions (Angermeier and Winston 1999; Jackson et al. 2001; Wenger et al. 2008). The occurrence of taxa or predictable assemblages is a result of

historical and regional processes such as speciation and dispersal (Rahel and Hubert 1991; Lyons 1996), local habitat suitability and physicochemical conditions (Taylor et al. 1993), and biotic interactions like predation and competition (Gilliam et al. 1993; Jackson et al. 2001; Wenger et al. 2008). Changes to fish assemblage structure can occur in response to environmental change in vulnerable aquatic ecosystems following the introduction of invasive species (Petesse & Petrere, 2012), degradation (e.g., ocean temperature change; Magurran et al., 2015), or policy changes (Gibson-Reinemer et al., 2017). As a result, changes in fish assemblages can act as reliable signals of environmental change in aquatic ecosystems.

The Upper Mississippi River (UMR) is a unique ecosystem with international recognition as an important wetland ecosystem that supports diverse and abundant wildlife and fish populations (Schramm Jr. 2017, Houser 2022). Long-term monitoring of fishes has yielded broad spatial comparisons of assemblage structure (Chick et al. 2005) as well as seasonal differences in assemblage structure (Barko et al. 2005). These investigations were conducted in the early 2000's and the quality of river habitats in the UMR have changed since that time (Houser 2022), which have likely influenced fish assemblages. Although shifts in annual estimates of species abundances and trophic guild biomass have been linked to increased water clarity and expanding aquatic vegetation (Giblin 2017, Burdis et al. 2020, Bouska 2020; Ickes et al. 2022), the extent to which fish assemblages have changed is not well understood. Habitat restoration in the UMR has often prioritized conditions that support recreationally valuable centrarchids, particularly during winter when such required habitat conditions are more limiting (Gent et al. 1995; Knights et al. 1995; Mooney et al. 2025). However, resource managers are increasingly interested in understanding habitat associations of a broader range of native fishes. Within the UMR, a better

understanding of how distinct fish assemblages are organized, how they have shifted over time, and the causes of these shifts can inform river management decisions.

Relatively few datasets and analytical approaches are readily available to provide direct insight into nonrandom environmental mechanisms that regulate individual species occurrence patterns and structure fish assemblages (Guisan and Zimmermann 2000; Filipe et al. 2002; Peres-Neto 2004). These measures of assessment are useful in identifying associations between a changing mosaic of species assemblage structure through time and their effects on single species occurrence (Angermeier and Winston 1999; Jackson et al. 2001; Olden et al. 2002). Clustering algorithms can define biological assemblages and shifts through time based on species composition to inform ecology and management. Defining biological assemblages helps researchers to delve deeper into environmental drivers that are specific to each assemblage (Szalóky et al., 2021; Delaney & Larson, 2022). Quantitative characterization of assemblages also helps managers define their management targets and evaluate the habitat conditions and trade-offs required to support those assemblages. In the UMR, clustering analyses have been useful to describe common floodplain forest tree assemblages important for forestry and floodplain ecology (Windmuller-Campione et al., 2022) and also for aquatic vegetation types important for managing wildlife habitat and achieving ecological restoration goals (Larson et al. 2023b). However, clustering analysis has not yet been applied to describe changes in the fish assemblages of the UMR, which could provide valuable insight for both valuable sport fish as well as ecologically important native species.

In this paper, we use 32 years of UMR fish monitoring data to (1) identify common fish assemblages and (2) evaluate how assemblage structure has changed across time and space. We applied clustering methods to characterize recurrent assemblages within four study reaches

spanning approximately 400 km of the UMR. We then explored whether assemblage structure shifted over time as fishes experienced substantial environmental change throughout the study period. The spatial and temporal patterns revealed here can help ecologists better understand how assemblages respond to environmental drivers and can assist river managers in refining and communicating fish management goals.

METHODS

Study area

The UMR flows approximately 2,200 kilometers through the central United States, where its 29 navigation locks and dams create a series of navigation pools, each defined by the discrete river segments between consecutive lock and dam structures. Installed in the late 1930s, these structures raised water levels and established shallow habitats that now support aquatic plant communities throughout the northern pools of the river.

This study focuses on four reaches spanning roughly 400 river kilometers: Upper Pool 4, Lower Pool 4, Pool 8, and Pool 13. These locations are part of the Long Term Resource Monitoring (LTRM) element of the Upper Mississippi River Restoration Program and were selected because they support 89 documented fish species (Ickes et al., 2022) and contain comprehensive long-term datasets on fishes, aquatic vegetation, water quality, hydrology, geomorphology, and land use (<https://umesc.usgs.gov/ltrm-home.html>). These reaches are distinguished by heterogeneous geomorphic structure with abundant backwaters, side channels, and open water areas (De Jager et al., 2018). They have undergone ecosystem regime shifts characterized by improved water quality and the resurgence of aquatic vegetation (Larson et al., 2023a; Larson et al., 2023b), along with a transition in fish community composition from

dominance by non-native common carp *Cyprinus carpio* to a more functionally diverse assemblage (Bouska, 2020).

Data collection and processing

We used fish community data collected by the LTRM. Since 1993, LTRM fish biologists have conducted annual sampling within in each study reach using multiple gears within a stratified random sampling design (detailed in Ratcliff et al., 2014). For this analysis, we restricted observations made to daytime electrofishing samples conducted each year between 16 September and 31 October. Each electrofishing sample consisted of a standardized 15-minute run along the shoreline. Captured fish were temporarily held in onboard live wells, identified to species, measured for maximum total length, and then released alive.

Statistics

We used the publicly available LTRM Fisheries Dataset, accessed on 1 November 2025 (<https://umesc.usgs.gov/ltrm-home.html>). All analysis scripts and derived datasets are publicly archived on GitHub (https://github.com/jppeirce/ltrm_fish). We retained daytime electrofishing samples (gear code D) collected during the autumn sampling period (period 3) from Pools 4, 8, and 13. We parsed Pool 4 into two separate reaches (Upper Pool 4 and Lower Pool 4) because of known differences in their biotic and abiotic characteristics. Samples affected by gear failure or site inaccessibility were removed, and records containing unidentified fishes and hybrids were excluded. To reduce data sparsity, we omitted species present in fewer than 5% of samples. The final dataset included 2,412 samples.

We identified assemblage-based fish clusters using hierarchical agglomerative clustering applied to presence–absence data aggregated at the site level. Pairwise compositional dissimilarity between assemblages was quantified with the Jaccard distance, consistent with

previous ecological applications involving binary occurrence data (Clarke et al. 2006). Prior to clustering, we evaluated the distribution of median pairwise Jaccard distances per assemblage to screen for extreme outliers; although a small number of potential outliers were apparent, all observations were retained to preserve the full range of observed variation. Hierarchical clustering was performed using the average-linkage algorithm on the Jaccard distance matrix, and the resulting dendrogram was cut at a distance threshold to generate cluster assignments.

To assess the robustness of the inferred cluster structure to sampling variation and choice of cut height, we conducted a bootstrap-based stability analysis across a grid of candidate dendrogram cut heights (Hennig 2007). For each threshold, we: (i) computed cluster assignments for the full dataset; (ii) repeatedly drew random subsamples of assemblages, re-clustered them, and (iii) measured concordance between subsample and full-data partitions using the adjusted Rand index (ARI) (Hubert & Arabie 1985). The final cut height was selected using multiple complementary criteria: maximizing bootstrap stability as quantified by the adjusted Rand index (ARI), minimizing local sensitivity between partitions at adjacent cut heights, limiting over-fragmentation by reducing the number of clusters containing fewer than 30 samples, and ensuring that no cluster exceeded 1,000 samples. This upper bound was imposed to prevent a single dominant cluster from obscuring finer-scale, ecologically meaningful assemblages.

The fish assemblage represented by each resulting cluster was characterized using various descriptors of the species composition. To guide cluster interpretation, we examined species prevalence within each cluster using heatmaps and summary tables. For each cluster, we summarized dominant species, overall species richness, species with relatively high prevalence, and absent species. We also incorporated an existing habitat guild classification scheme (WEST

Consulting 2000) and trophic guild classification (Becker 1983, O'Hara et al. 2007) to provide additional ecological context for assemblage descriptions.

To assess temporal trends in fish assemblage composition, we calculated the annual proportion of samples assigned to each cluster within each study reach (Upper Pool 4, Lower Pool 4, Pool 8, and Pool 13). We used segmented (piecewise) linear regression to detect potential breakpoints in these temporal patterns. To evaluate whether the segmented model provided a significantly better fit than a simple linear model, we performed an F-test for nested models, a common approach for identifying regime shifts in ecological time series (Anderson, 2009; Zeileis, 2003). All analyses were implemented in Python using *SciPy* (Virtanen, 2000). We tested whether and where statistical 'breakpoints' occurred in the 30-year time series that could infer assemblage shifts in time. Segmented regression was applied to each time series for all pool-cluster combinations. Initial fits using simple linear models were compared to segmented models, and model improvements ranged from 3% to 66%, depending on the study reach. Based on these results, we reported and mapped the segmented regression results that included one statistically significant ($p < 0.05$) breakpoint per pool-cluster combination.

RESULTS

Clustering revealed eight key assemblages

Cutting the hierarchical clustering dendrogram at a distance threshold of 0.718 yielded 27 clusters (mean ARI = 0.441 ± 0.051). We focused on eight primary clusters with at least 30 samples each, which together accounted for approximately 91% of all observations. The largest cluster comprised about 40% of total samples, second largest 19%, and each of the remaining six clusters less than 15.5% (Figure 2).

Species composition within these cluster was complex but interpretable based on the prevalence heatmap and cluster summary tables (Figure 2). Across all samples, 40 species were represented, spanning limnophilic to rheophilic habitat guilds and including a wide range of trophic guilds (Table 1). Largemouth Bass and Bluegill occurred in the greatest number of samples. Among the eight primary clusters, Largemouth Bass, Bluegill, Shorthead Redhorse and Gizzard Shad were the most ubiquitous species, while Emerald Shiner, Common Carp, and Yellow Perch each exceeded 50% prevalence in at least one cluster.

The eight statistically derived clusters were ecologically interpretable and are summarized below.

- Cluster 1 represented a diverse, functionally complex assemblage in which all 40 species occurred in at least 2% of samples. It exhibited high prevalence of Bluegill, Largemouth Bass, Emerald Shiner, and Shorthead Redhorse, with additional high prevalence (>50% of samples) of Smallmouth Bass, Spotfin Shiner, Bullhead Minnow, Common Carp, Yellow Perch, and Rock Bass. Although Johnny Darter and Pugnose Minnow were relatively uncommon overall, they were more prevalent here than in other clusters.
- Cluster 2 represented a diverse, predator-rich, limnophilic assemblage dominated by Bluegill, Largemouth Bass, and Yellow Perch. Northern Pike, Pumpkinseed, Spotted Sucker, and Black Crappie were also frequent (>50%). Golden Shiner and Bowfin occurred disproportionately in this cluster, whereas Channel Catfish, White Bass, Freshwater Drum, and Shorthead Redhorse were comparatively rare.
- Cluster 3 was also diverse, dominated by planktivores and detritivores, including Gizzard Shad, Common Carp, and Emerald Shiner along with common occurrences of Largemouth Bass and Bluegill. Several species that were infrequent overall-such as

Orangespotted Sunfish, Smallmouth Buffalo, White Crappie, River Carpsucker, and Bigmouth Buffalo-were relatively common in this cluster.

- Cluster 4 was characterized by Gizzard Shad, Shorthead Redhorse, Common Carp and Emerald Shiner, but also included high prevalence of piscivorous species more adapted to riverine (lotic flow) habitats, such as Smallmouth Bass, White Bass, Sauger, and Walleye.
- Cluster 5 was dominated by several species of Redhorse (Shorthead Redhorse, Silver Redhorse, River Redhorse, and Golden Redhorse) as well as Smallmouth Bass that favor lotic habitat.
- Clusters 6, 7 and 8 were more simplified assemblages dominated by only a few species. These clusters differed primarily in habitat affinities, representing of pelagic/benthic (Cluster 6), riverine (Cluster 7) and limnophilic (Cluster 8) community types.

Spatial patterns varied among river reaches

The distribution of fish assemblage clusters differed notably among the four study reaches (Figure 4). Cluster 1, which was the most taxonomically and functionally diverse, was dominant in Pool 8 (approximately 61% of samples) and was also common in Lower Pool 4 (approximately 38%). Cluster 2, a diverse collection characterized by a high prevalence of top predators, occurred frequently in Lower Pool 4, Pool 8 and Pool 13, but was nearly absent from Upper Pool 4. Cluster 3, which included abundant planktivores and detritivores, was most common in Pool 13 and Upper Pool 4 and occurred only rarely in Lower Pool 4 and Pool 8. Cluster 4, dominated by piscivorous, riverine fishes, was found almost exclusively in Upper Pool 4. Clusters 5 through 8 were either absent or at low prevalence (<9%) across all study reaches.

Cluster distributions also varied by habitat classes and were indicative of broad habitat preferences (Figure 5). Clusters 1 and 3 were ubiquitous across the main channel borders (structured and unstructured), side and tertiary channels, and floodplain lakes. Clusters 2 and 8 occurred most frequently in contiguous floodplain lakes and tertiary channels. Clusters 4 and 6 were most common in the main channel habitats (structured and unstructured) and side channels. Cluster 5 was primarily associated with structured main channel borders. Cluster 7 occurred most frequently in floodplain lakes.

Temporal shifts of fish assemblages over 30 years

Segmented regression identified statistically significant temporal shifts in the relative prevalence of fish assemblage clusters across all study reaches throughout the 30-year study period. Pool-wide breakpoint timing and directionality are summarized spatially in Figure 5 and Figure 6. Notable patterns are described below.

- In Upper Pool 4, a significant breakpoint in 1998 ($F = 9.08$, $p = 0.0013$) marked a reversal in Cluster 3 prevalence from a declining to an increasing trend, reflecting a growing dominance of the planktivore- and detritivore-dominated assemblage in later years.
- In Lower Pool 4, significant temporal shifts were detected in both Clusters 2 and 4. Cluster 2 prevalence approximately quadrupled between 1993 and 2009 ($F = 3.89$, $p = 0.0328$; Figure 5). A similar increasing trend in Cluster 2 was observed in Pools 8 and 13, though breakpoints in those reaches did not reach statistical significance. Cluster 4 exhibited the opposite trajectory: a steep decline in prevalence between 1993 and 1996 transitioned to a more gradual decline in subsequent years, a shift captured by a highly significant breakpoint ($F = 319.01$, $p < 0.001$).

- In Pool 8, Cluster 1 prevalence increased approximately three-fold between 1993 and 1995 before giving way to slight declines in later years, corresponding to a significant breakpoint ($F = 18.71$, $p < 0.001$).
- In Pool 13, Cluster 3 initially increased between 1993 and 1997 before declining sharply following a significant breakpoint in 1997 ($F = 4.80$, $p = 0.0161$). Cluster 4 exhibited a significant breakpoint in 1995 ($F = 15.63$, $p < 0.001$), within one year of an analogous shift observed in Lower Pool 4, suggesting a degree of synchrony in assemblage dynamics across these reaches.

Taken together, these temporal patterns are consistent with the broader shifts in fish assemblage composition observed across aquatic areas throughout the 30-year study period (Figure 7).

DISCUSSION

Our analysis identified eight ecologically meaningful fish assemblages and demonstrated substantial reorganization of assemblage structure over the 32 years of fish monitoring. The temporal shifts in fish assemblage prevalence align with documented system-wide changes, including improved water quality, recovery of aquatic vegetation, and restoration of habitat. Collectively, these results address a gap in our understanding of how fish assemblages respond to long-term environmental change and provide an updated perspective on assemblage structure that can inform habitat management and conservation.

The spatial distribution of clusters reflects habitat preferences and trophic gradients across the major habitat types of the UMR (e.g., main-channel borders, side channels, and floodplain lakes). As expected, cluster prevalence was uneven among and within pools. Upper Pool 4 was especially distinct, exhibiting high interannual variability in assemblage composition.

In contrast, Lower Pool 4 and Pool 8 showed strong commonalities, with elevated prevalence of diverse, functionally complex and predator-rich, limnophilic assemblages (Figure 5). Temporal shifts were also evident within pools; for example, in Pool 13 there was a transition from dominance by planktivores and detritivores in earlier years toward a more diverse and functionally complex composition.

The two diverse and functionally complex assemblages (Clusters 1 and 3) were widely distributed across different types of habitats, likely representing shifting habitat use with changing stage and temperature (Gutreuter et al., 2010). Both of these assemblages exhibited strong increases early in the period of record and coincide with system-wide improvements in water quality (Wiener & Sandheinrich, 2010), increases in invertebrate prey (Fremling et al., 1989a; Fremling et al., 1989b), and reductions in common carp abundance across the three study reaches (Bouska 2020). Similar findings have been reported for the Illinois River and attributed to reductions in point source contributions following the implementation of the Clean Water Act (Gibson-Reinemer et al., 2017). The prevalence of these two assemblages differed among pools and likely represent differences in water clarity. Specifically, several species more prevalent in Cluster 1 such as smallmouth bass, rock bass, spotfin shiner, weed shiner, bullhead minnow, and logperch favor clearer conditions whereas several species more prevalent in Cluster 3 are more tolerant of turbid conditions, including gizzard shad, channel catfish, freshwater drum, bigmouth buffalo, and smallmouth buffalo (Becker 1983; Pflieger 1997).

An increased prevalence of the diverse, predator-rich, limnophilic assemblage (Cluster 2) in Lower Pool 4, Pool 8 and Pool 13 align with clearer water and recovered submersed vegetation. These patterns are consistent with evidence that reduced total suspended solids and increased macrophytes favor visual predators such as largemouth bass, bluegill, and northern

pike (Giblin 2017). Further, declines in turbidity have shifted backwater fish assemblages toward vegetation-associated taxa (Burdis et al., 2020). Increased prevalence of this fish assemblage is also consistent with trends in piscivore biomass across these reaches (Bouska, 2020). The diverse, predator-rich, limnophilic assemblage was associated with contiguous floodplain lakes, where water clarity changes have been linked to a combination of factors including reduced common carp biomass, increased aquatic vegetation cover, and lower total phosphorus concentrations (Carhart et al. 2024). Higher turbidity and hydrologic variability constrain persistent submersed aquatic vegetation recovery and limit the prevalence assemblage of this assemblage in Upper Pool 4 (Jankowski et al. 2023; Larson et al. 2023b).

Other fish assemblages appear to reflect specialized habitat use in addition to broader environmental gradients. In particular, the riverine piscivore (Cluster 4) and riverine redhorse assemblages were more prevalent in rheophilic habitats, such as main channel borders and side channels. The riverine piscivore assemblage had a greater prevalence Upper Pool 4 and Pool 13, whereas the riverine redhorse assemblage was most prevalent in Lower Pool 4 and Pool 8. Several redhorses are recognized to be moderately intolerant of turbidity (Trebitz et al. 2007), which may explain pool-to-pool differences in the prevalence of rheophilic fish assemblages.

Our approach effectively delineated fish assemblage structure and identified shifts over time. In the future, incorporating abundance or biomass into the clustering framework would allow trends to reflect population change more directly than presence alone. In addition, quantifying the linkages between assemblage shifts and environmental drivers would help distinguish slow or lagged, system-level demographic responses from site-specific habitat effects, clarifying mechanisms behind observed patterns. Finally, expanding analyses to additional reaches and seasons could improve generality and reveal season-dependent

co-occurrences: we would expect seasonal shifts in habitat use resulting from different life-history needs and changing interannual environmental conditions (Galat and Zweimüller 2001; Gutreuter et al. 2010, Gillette et al. 2006). Including additional seasons and reaches would sharpen inference about when and where assemblages reorganize.

Large river ecosystems contain strong environmental gradients such as water depth, velocity, and water quality that are widely recognized as key drivers of fish assemblage structure (Fremling et al. 1989; Chea et al. 2016). Because many riverine species depend on specific habitat characteristics, these gradients shape where species occur. At the same time, habitat use of riverine fishes is not static: the distribution of fishes can shift based on life-history needs (e.g., spawning; Galat and Zweimüller 2001) and in response to variable discharge and temperature conditions (Gillette et al. 2006; Moore et al. 2022). These interacting factors make it difficult to delineate fish assemblages in dynamic river systems (Eros et al. 2017). Although past approaches have successfully identified longitudinal patterns in assemblage organization (Huet 1959; Chea et al. 2016; Smith et al. 2016), future work will explore how environmental gradients and temporal variability jointly influence assemblage structure.

Conclusion

Our analysis of long-term fish monitoring data in the UMR revealed shifts in fish assemblage structures over the past three decades. Three dominant assemblages emerged, each reflecting distinct ecological preferences and habitat associations. Notably, there was a regional trend away from flowing-water species toward limnophilic and small-bodied assemblages, particularly in off-channel backwaters. These changes likely reflect broader environmental improvements (like increases in water quality, water quantity, and aquatic vegetation) and habitat

restoration efforts. These findings highlight the dynamic nature of riverine fish assemblages and underscore the importance of long-term data for informing habitat restoration and management.

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537 TABLES

538 **Table 1.** A list of species names, codes, and habitat preferences for all 40 fishes included in our
 539 fish assemblage analysis. Habitat and trophic guild descriptions were derived from Becker 1983,
 540 WEST Consultants 2000, and O’Hara et al. 2007.

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Species code	Common name	Scientific name	Habitat guild	Trophic guild
BHMW	Bullhead minnow	<i>Pimephales vigilax</i>	Rheo-limnophilic	Invertivore/ Herbivore
BKCP	Black crappie	<i>Pomoxis nigromaculatus</i>	Limnophilic	Invertivore/ Piscivore
BKSS	Brook silverside	<i>Labidesthes sicculus</i>	Rheo-limnophilic	Planktivore/ Invertivore
BLGL	Bluegill	<i>Lepomis macrochirus</i>	Limnophilic	Invertivore
BMBF	Bigmouth buffalo	<i>Ictiobus cyprinellus</i>	Pelagic limno-rheophilic	Invertivore
BWFN	Bowfin	<i>Amia calva</i>	Limnophilic	Piscivore
CARP	Common carp	<i>Cyprinus carpio</i>	Limno-rheophilic	Invertivore/ Detritivore
CNCF	Channel catfish	<i>Ictalurus punctatus</i>	Rheophilic	Invertivore/ Carnivore
ERSN	Emerald shiner	<i>Notropis atherinoides</i>	Rheo-limnophilic	Planktivore
FWDM	Freshwater drum	<i>Aplodinotus grunniens</i>	Limno-rheophilic	Invertivore/ Molluscivore
GDRH	Golden redhorse	<i>Moxostoma erythrurum</i>	Limno-rheophilic	Invertivore
GDSN	Golden shiner	<i>Notemigonus crysoleucas</i>	Limno-rheophilic	Invertivore/ Herbivore
GNSF	Green sunfish	<i>Lepomis cyanellus</i>	Limnophilic	Invertivore/ Carnivore
GZSD	Gizzard shad	<i>Dorosoma cepedianum</i>	Limnophilic	Planktivore
JYDR	Johnny darter	<i>Etheostoma nigrum</i>	Limno-rheophilic	Invertivore
LGPH	Logperch	<i>Percina caprodes</i>	Limno-rheophilic	Invertivore
LMBS	Largemouth bass	<i>Micropterus salmoides</i>	Limnophilic	Invertivore/ Piscivore
MMSN	Mimic shiner	<i>Notropis volucellus</i>	Rheo-limnophilic	Invertivore/ Herbivore

NTPK	Northern pike	<i>Esox lucius</i>	Limnophilic	Piscivore
OSSF	Orangespottedsunfish	<i>Lepomis humilis</i>	Limnophilic	Invertivore
PGMW	Pugnoseminnow	<i>Opsopoeodus emiliae</i>	Limnophilic	Detritivore
PNSD	Pumpkinseed	<i>Lepomis gibbosus</i>	Limnophilic	Invertivore/ Molluscivore
QLBK	Quillback	<i>Carpionodes cyprinus</i>	Limno- rheophilic	Invertivore/ Detritivore
RKBS	Rock bass	<i>Ambloplites rupestris</i>	Limno- rheophilic	Invertivore/ Piscivore
RVCS	River carpsucker	<i>Carpionodes carpio</i>	Limno- rheophilic	Planktivore/ Detritivore
RVRH	River redhorse	<i>Moxostoma carinatum</i>	Rheo- limnophilic	Invertivore
RVSN	River shiner	<i>Notropis blennioides</i>	Rheo- limnophilic	Invertivore
SFSN	Spotfin shiner	<i>Cyprinella spiloptera</i>	Rheo- limnophilic	Invertivore/ Detritivore
SGER	Sauger	<i>Sander canadense</i>	Rheo- limnophilic	Invertivore/ Piscivore
SHRH	Shorthead redhorse	<i>Moxostomamacrolepidotum</i>	Rheo- limnophilic	Invertivore
SMBF	Smallmouth buffalo	<i>Ictiobus bubalus</i>	Pelagic limno- rheophilic	Invertivore/ Herbivore
SMBS	Smallmouth bass	<i>Micropterus dolomieu</i>	Limno- rheophilic	Invertivore/ Piscivore
SPSK	Spotted sucker	<i>Minytrema melanops</i>	Limno- rheophilic	Invertivore
STSN	Spottail shiner	<i>Notropis hudsonius</i>	Limno- rheophilic	Invertivore/ Planktivore
SVRH	Silver redhorse	<i>Moxostoma anisurum</i>	Limno- rheophilic	Invertivore
WDSN	Weed shiner	<i>Notropis texanus</i>	Limno- rheophilic	Detritivore
WLYE	Walleye	<i>Sander vitreum</i>	Limno- rheophilic	Invertivore/ Piscivore
WTBS	White bass	<i>Morone chrysops</i>	Pelagic limno- rheophilic	Invertivore/ Piscivore
WTCP	White crappie	<i>Pomoxis annularis</i>	Limnophilic	Invertivore/ Piscivore
YWPH	Yellow perch	<i>Perca flavescens</i>	Limno- rheophilic	Invertivore/ Piscivore

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FIGURES

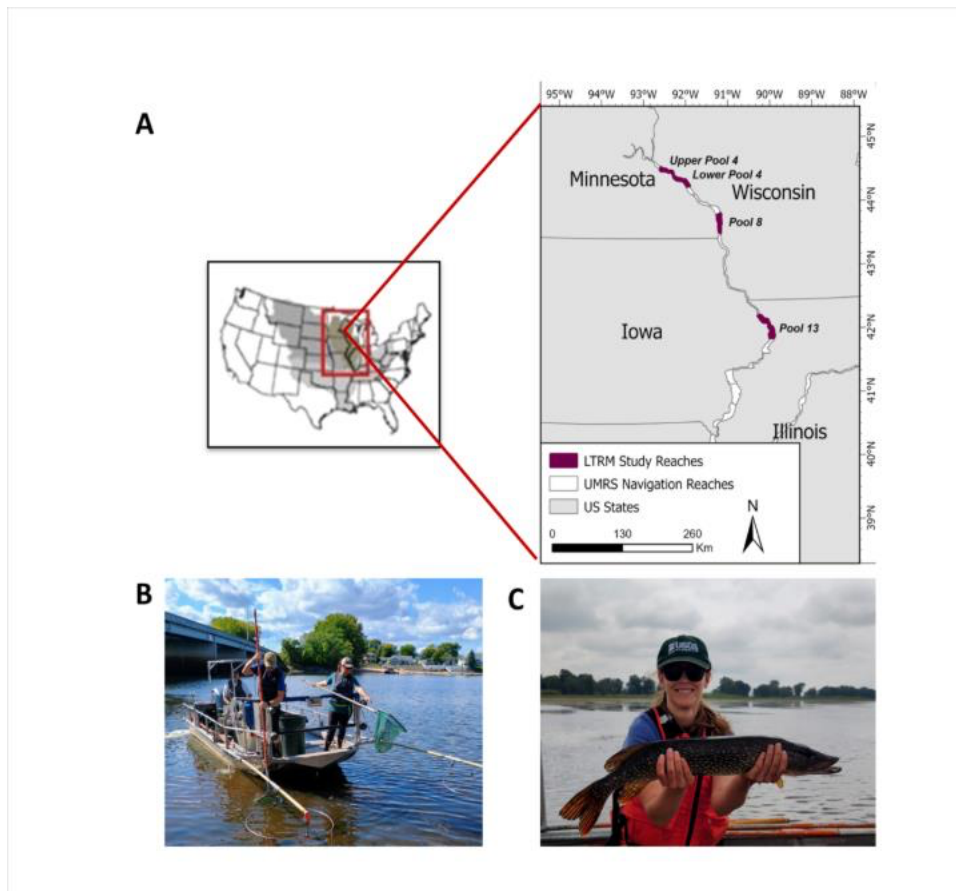


Figure 1. Map of the United States of America with the Mississippi River basin (shaded in gray), and then the inset shows the long term resource monitoring (LTRM) study areas (four maroon areas in panel A). Fishes were collected with standardized daytime electrofishing protocols (panel B) and then speciated and measured for length individually (panel C). Map and pictures by author Danelle Larson, used with permission.

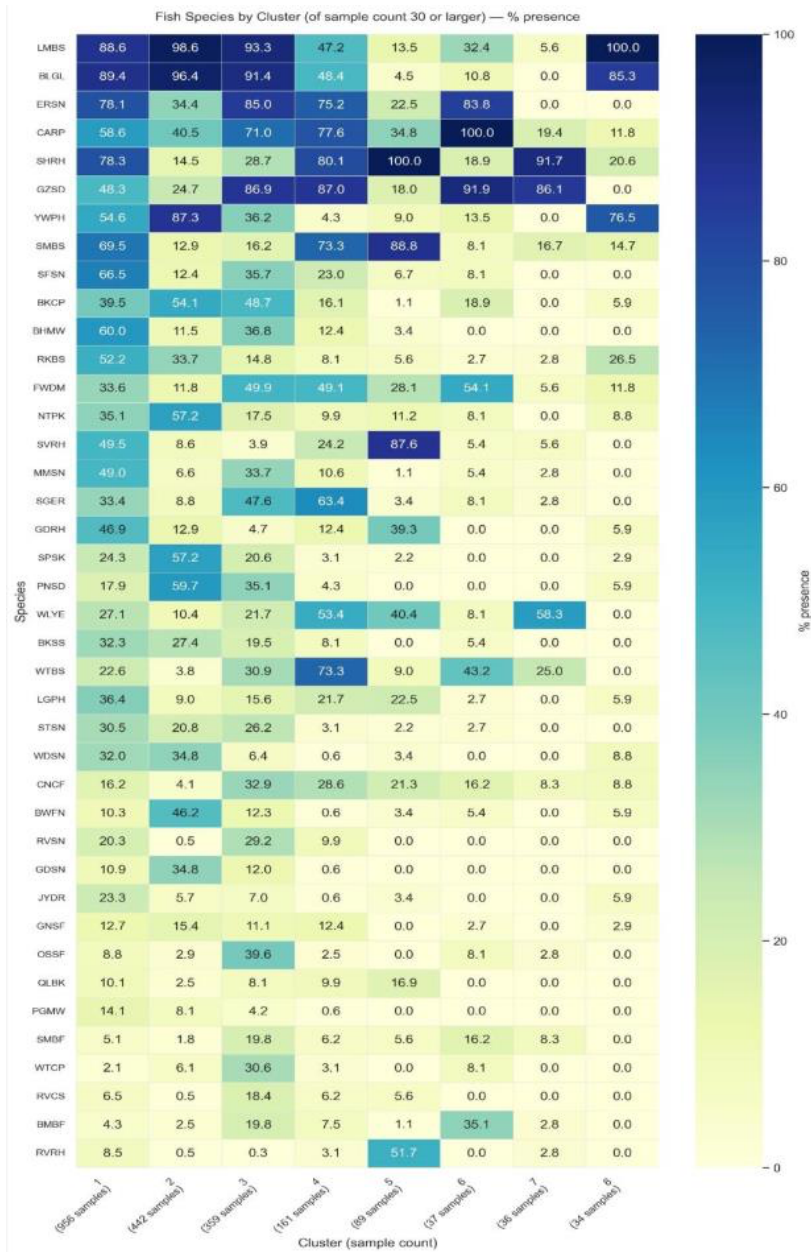
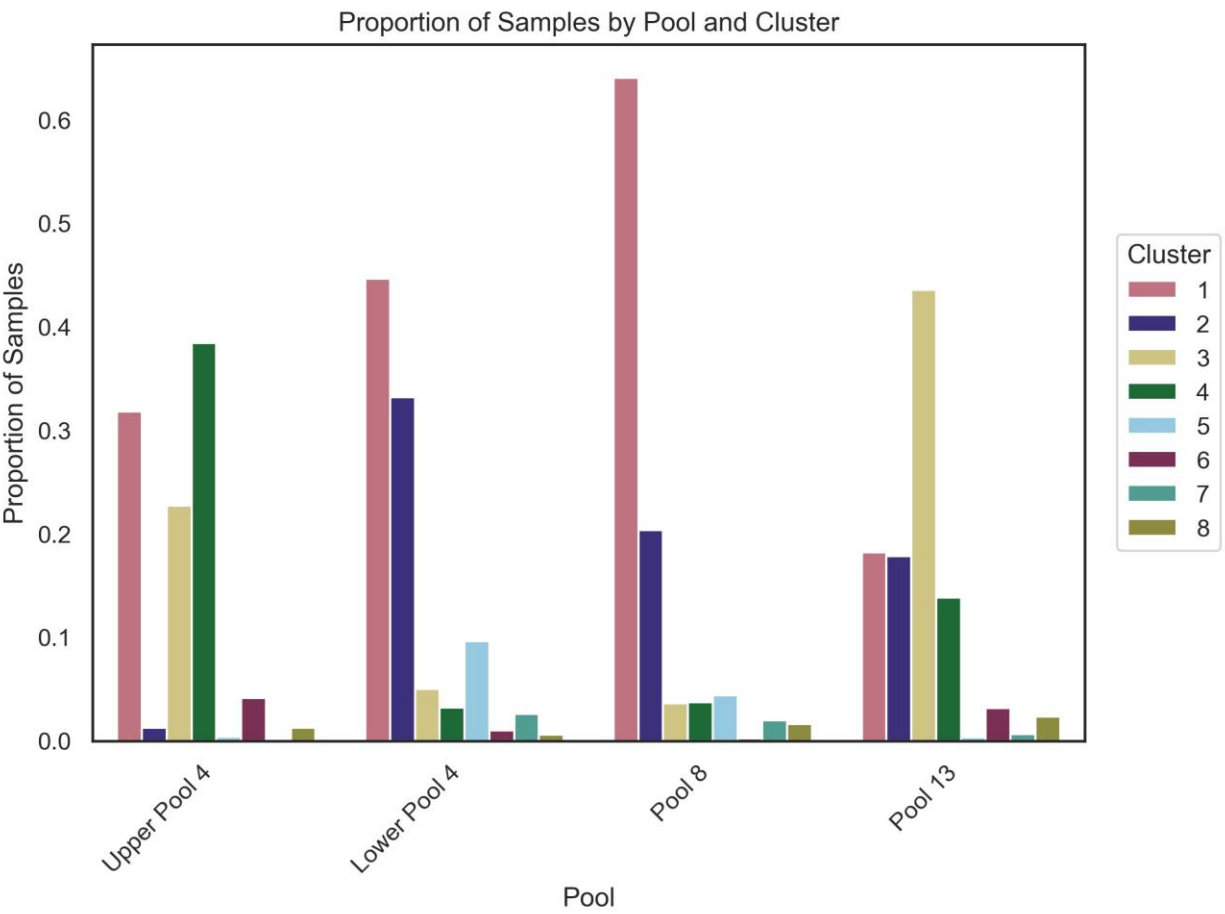


Figure 2. Fish species codes (full species names in Table 1) for eight clusters, whereby cluster is defined by fish assemblage composition. The heatmap shows the percentage of sampling sites where each species was present within each cluster. Cell values and color represent prevalence (0–100%), with darker blues indicating higher prevalence of that fish species. Clusters are ordered by sample size or the number of sites classified within that cluster (shown in parentheses).

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Figure 3. The proportion of samples belonging to distinct fish assemblage clusters across four

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Upper Mississippi River study reaches.

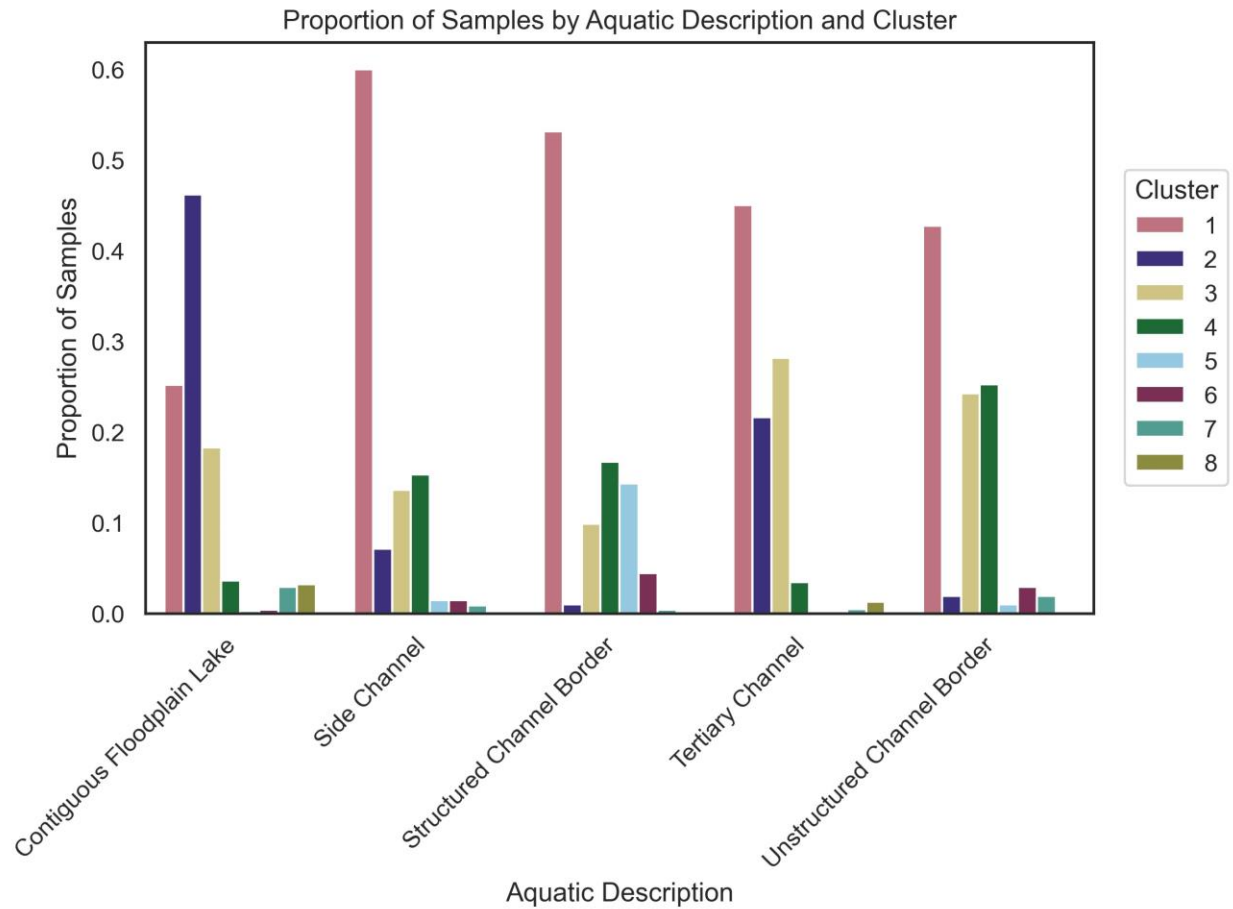


Figure 4. The proportion of samples belonging to distinct fish assemblage clusters across five Upper Mississippi River aquatic descriptions.

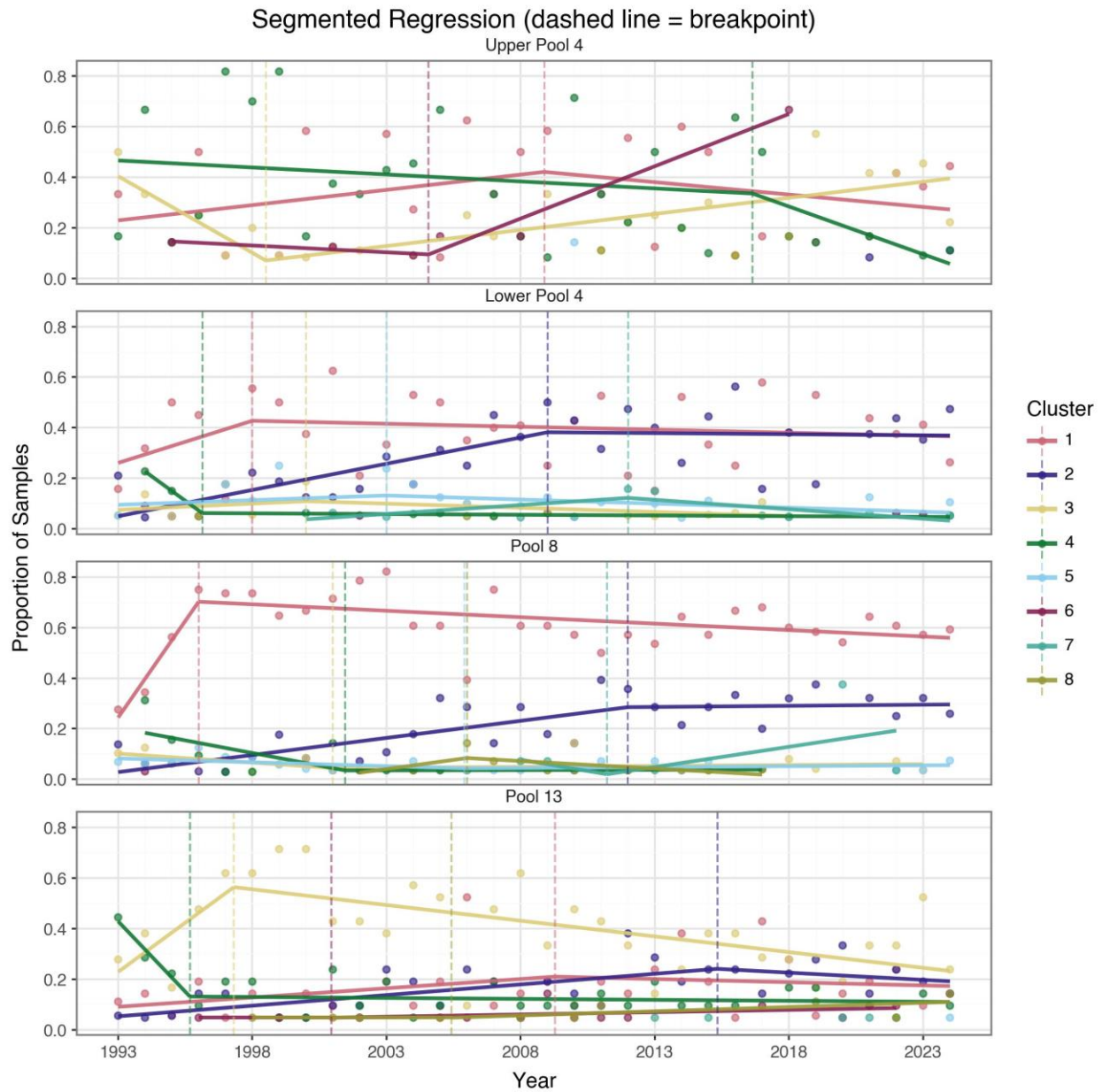


Figure 5. Temporal trajectories (1993–2024) of fish assemblage cluster proportions across four Upper Mississippi River study areas. Points represent observed proportions of sampling sites assigned to each cluster (colored by cluster ID) within each year. Solid lines show fitted segmented regression models with one breakpoint, and dashed vertical lines indicate the estimated breakpoint year for each study reach-cluster combination.

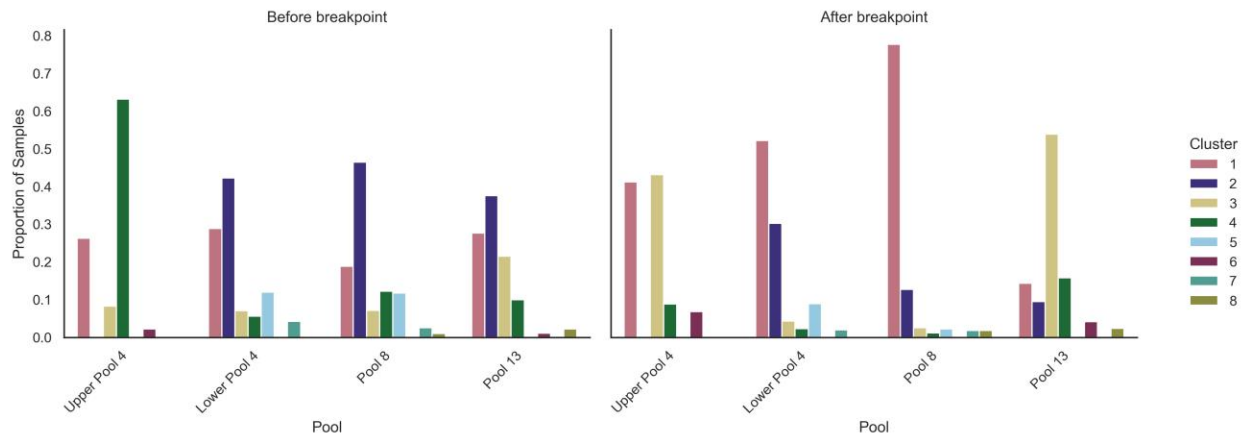


Figure 6. Cluster composition by pool before and after estimated breakpoints. Bars show the proportion of samples assigned to each fish assemblage cluster (clusters 1–8) within each pool and period (Before breakpoint vs After breakpoint). Comparisons indicate pool-specific shifts in relative cluster representation over time.

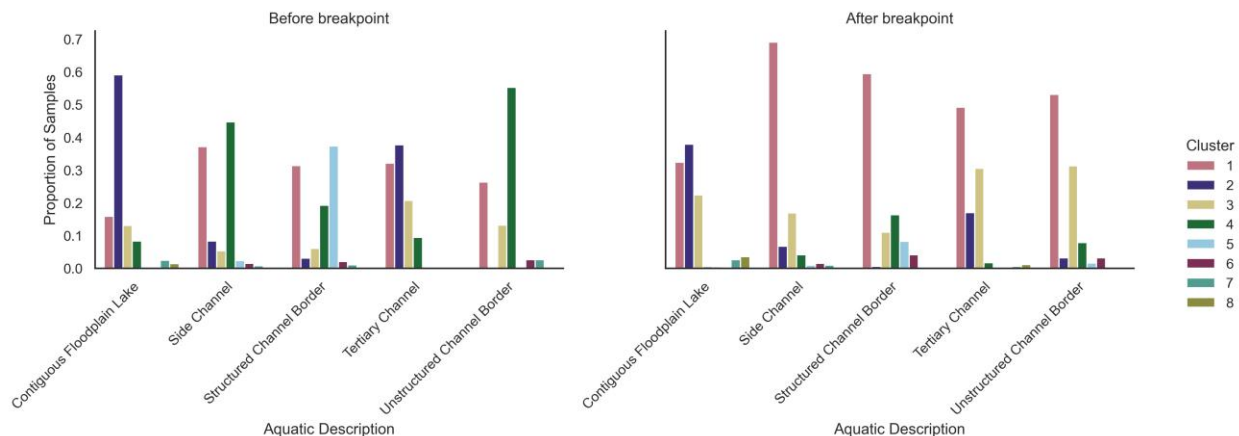


Figure 7. Cluster composition by aquatic description before and after estimated breakpoints.

Bars show the proportion of samples assigned to each fish assemblage cluster (clusters 1–8) within each aquatic description and period (Before breakpoint vs After breakpoint). Comparisons indicate aquatic description-specific shifts in relative cluster representation over time.

Extra (?)

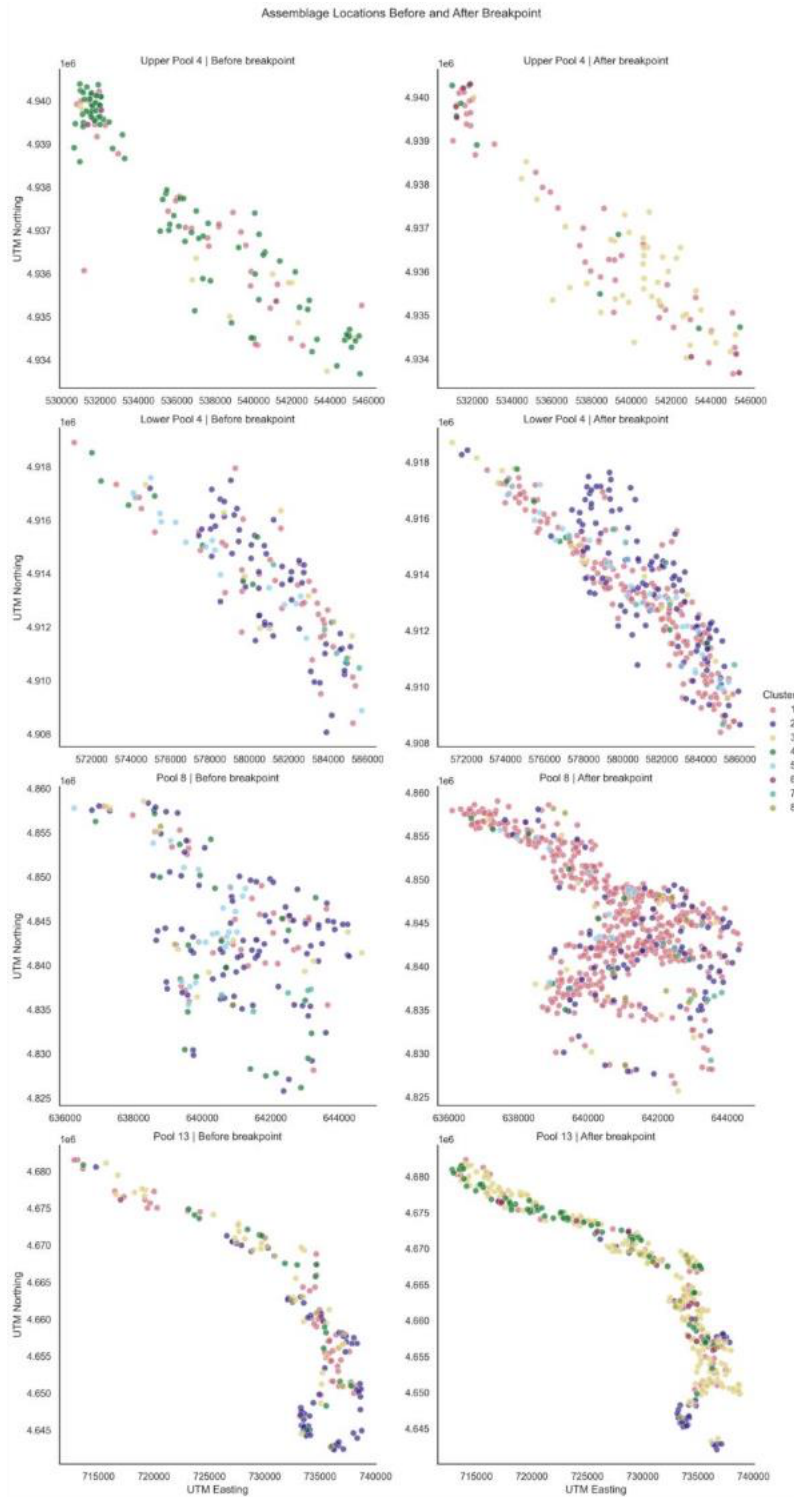


Figure 5. Spatial distribution of fish assemblage clusters before and after estimated temporal breakpoints across four Upper Mississippi River study reaches. Points represent sampling

591 locations in UTM coordinates; colors denote cluster ID (1 = gray, 2 = green, 3 = blue, 4 = red).
592 Rows correspond to pools (Upper Pool 4, Lower Pool 4, Pool 8, Pool 13), and columns separate
593 time before and after the breakpoint year inferred from segmented regressions of annual cluster
594 proportions. Statistically significant breakpoints occurred in 1998 for Upper Pool 4, 2009 for
595 Lower Pool 4, 1995 for Pool 8, and 1997 for Pool 13. Plots highlight changes in the spatial
596 footprint of dominant assemblage types across time periods.

597

?NEEDED?

Table 2. The percentage of sites (includes average and variance) according to Mississippi River USA study areas (gray-scaled columns) that are classified into various clusters, as determined by a hierarchical agglomerative clustering algorithm. ‘NaN’ indicates the cluster was not present within a study area.

Cluster	Upper Pool 4 avg	Upper Pool 4 var	Lower Pool 4 avg	Lower Pool 4 var	Pool 8 avg	Pool 8 var	Pool 13 avg	Pool 13 var
1	28%	5%	38%	2%	61%	1%	16%	1%
2	1%	0%	28%	2%	19%	1%	16%	1%
3	19%	2%	4%	0%	3%	0%	38%	3%
4	27%	5%	2%	0%	2%	0%	8%	0%
5	0%	0%	8%	0%	4%	0%	0%	0%
6	6%	1%	0%	0%	NaN	NaN	3%	0%
7	4%	2%	1%	0%	0%	0%	3%	0%
8	NaN	NaN	2%	0%	2%	0%	1%	0%

SUPPLEMENTAL MATERIALS

Supplement 1. Diagnostics of hierarchical clustering to assure stability of thresholds.

```

612 =====
613 ==
614 RESULTS: Most Stable Thresholds
615 =====
616 ==
617
618 1. HIGHEST STABILITY (Mean ARI):
619     Threshold: 0.810
620     Mean ARI: 0.6691 ± 0.0797
621     Number of clusters: 23
622     Large clusters (≥10): 3
623
624 2. MOST CONSISTENT (Lowest CV of ARI):
625     Threshold: 0.740
626     CV of ARI: 0.0940
627     Mean ARI: 0.5166 ± 0.0486
628     Number of clusters: 44
629
630 3. BEST OVERALL BALANCE:
631     Threshold: 0.700
632     Mean ARI: 0.4557 ± 0.0492
633     CV of ARI: 0.1080
634     Number of clusters: 70
635     Large clusters (≥10): 8
636
637
638
639

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642 RECOMMENDATION:

643 Use threshold = 0.700 for maximum stability

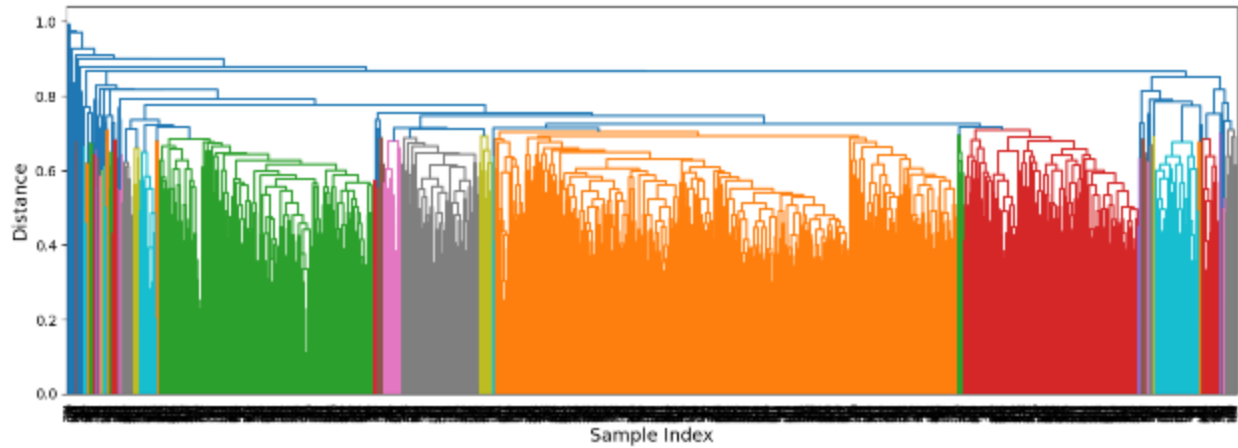
644 This produces 8 large clusters with

645 Mean ARI = 0.4557 (closer to 1.0 = more stable)

646

647 Supplement 2. Cladogram

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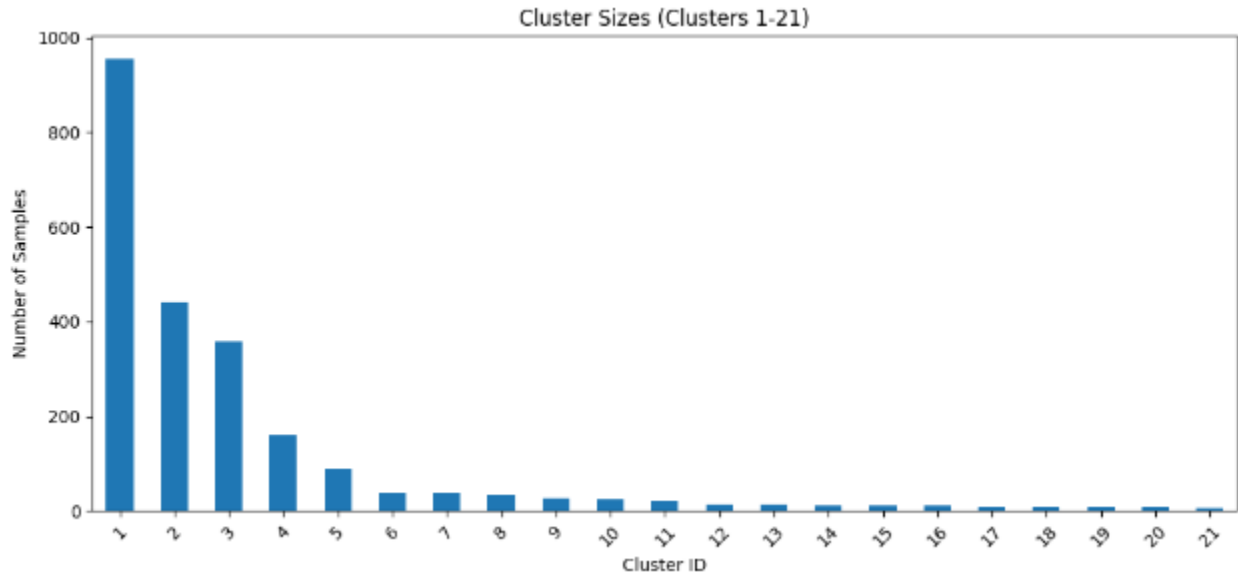


=== CLUSTER SUMMARY (threshold=0.71) ===

Total clusters: 65

Clusters with >10 samples: 16

Clusters with >50 samples: 5



Supplemental Table 1. List of fish species selected for the limnophilic (lentic) fish indicator in Ickes (2022) and their percentage contribution to total mass of limnophilic fishes sampled from all study reaches by daytime electrofishing from 1993 to 2019.

Common name	Scientific name	Total mass (percent)
Largemouth bass	<i>Micropterus salmoides</i>	40.17
Bluegill	<i>Lepomis macrochirus</i>	14.44
Bowfin	<i>Amia calva</i>	14.22
Weed shiner	<i>Notropis texanus</i>	7.04
Northern pike	<i>Esox lucius</i>	5.82
Black crappie	<i>Pomoxis nigromaculatus</i>	5.14
Longnose gar	<i>Lepisosteus osseus</i>	4.29
White crappie	<i>Pomoxis annularis</i>	3.07
Yellow perch	<i>Perca flavescens</i>	2.63
Golden shiner	<i>Notemigonus crysoleucas</i>	0.73
Spotted gar	<i>Lepisosteus oculatus</i>	0.61
Green sunfish	<i>Lepomis cyanellus</i>	0.43
Pumpkinseed	<i>Lepomis gibbosus</i>	0.38
Orangespotted sunfish	<i>Lepomis humilis</i>	0.19

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